

RANJITH PRASATH M V

Data Science Master’s Student | Python & Data Systems | GenAI & Sustainable Computing

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LinkedIn | GitHub | Portfolio

PROFILE

Master’s student in Data Science at UCLouvain with a Computer Science background and a growing interest in **applied AI, data systems, automation, and sustainable computing**. My main hands-on experience comes from building a large financial analytics and reconciliation project using Python, PostgreSQL, Flask, Docker, and data-processing pipelines. Within this project, I have also experimented with **RAG, ChromaDB-based retrieval, LLM-assisted parsing, and simple LLM-driven workflows**.

My Master’s thesis focuses on the **environmental impacts of different Generative AI user profiles**, with the goal of understanding how AI usage, computational workloads, infrastructure, energy consumption, water use, and environmental impact are connected. I am seeking an internship where I can strengthen my applied AI and research skills through experimentation, prototyping, testing, and real engineering problems.

EDUCATION

Université catholique de Louvain (UCLouvain)
Master [120] in Data Science: Information Technology

Louvain-la-Neuve, Belgium
Sep 2025 – Present

Master’s Thesis: *Evaluation of the Environmental Impacts of Different GenAI User Profiles*

- Investigating how different patterns of Generative AI use translate into computational workloads and environmental impacts.
- Studying indicators including greenhouse-gas impact, primary energy demand, water use, and resource consumption.
- Exploring the relationship between AI inference, infrastructure efficiency, workload characteristics, and sustainable computing.

Relevant coursework: Machine Learning, Databases, Data Mining, Statistics, Algorithms in Data Science.

Karpagam College of Engineering
B.E. Computer Science & Design
Final grade: **81.9%**

Coimbatore, India
Sep 2021 – May 2025

TECHNICAL SKILLS

Programming	Python, SQL
Data & Backend	PostgreSQL, pandas, NumPy, Flask, SQLAlchemy, REST APIs, data processing and reconciliation pipelines
GenAI Exposure	RAG, ChromaDB, LLM APIs, prompt-based workflows, LLM-assisted parsing, basic agent workflows
Engineering Tools	Docker, Docker Compose, Git, GitHub, pytest
Analytics	Fuzzy matching, anomaly detection, statistical analysis, Monte Carlo simulation, Power BI, Excel
Currently Learning	AI engineering, LangChain, LangGraph, MCP, AI automation, LLM inference and AI infrastructure

SELECTED PROJECT

RevenU – Financial Analytics & Reconciliation Platform GitHub

- Developed a Python-based financial analytics and reconciliation platform spanning multiple enterprise data domains including accounts receivable, accounts payable, banking, general ledger, treasury, revenue, inventory, and order data.
- Built rule-based and fuzzy-matching reconciliation workflows with confidence scoring, exception handling, escalation logic, and audit trails.
- Tested reconciliation workflows on datasets ranging from tens of thousands of records to synthetic stress tests of up to **500K rows**.
- Integrated experimental GenAI components including **ChromaDB-backed RAG memory**, LLM-assisted parsing of unstructured inputs, and LLM-generated explanations and action suggestions.
- Used PostgreSQL, Flask, SQLAlchemy, Docker, Redis/Celery components, and a React frontend to connect data processing, backend services, and user-facing workflows.

EXPERIENCE

Digital Marketing Executive Intern
Virtuospark Technologies Pvt Ltd

India
Aug 2024 – Apr 2025

- Built weekly and monthly analytical dashboards tracking revenue, ROI, costs, and operational KPIs across multiple channels.

- Used A/B testing and performance analysis to support budget-allocation and efficiency decisions.

SEO Analyst InternIndia

DOT Mappers IT Pvt LtdJan 2024 – Jul 2024

- Analysed performance data across more than **700 web pages** to identify patterns, anomalies, and optimisation opportunities.

RESEARCH & TECHNICAL INTERESTS

Applied AI • Generative AI • RAG • AI Automation • LLM Inference • AI Infrastructure • Efficient AI • Sustainable Computing

Languages: English – Fluent (IELTS 7) | French – A2 | Tamil – Native